



IVI White Paper 2026

Lean Product Lifecycle Management for Manufacturing Innovation

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EXECUTIVE SUMMARY

This white paper proposes “Lean Product Lifecycle Management (Lean PLM)” as a next-generation approach to manufacturing in an era of increasing uncertainty. Three key visions—sustainability, inclusiveness, and autonomy—and outlines four future factory models: connected, distributed, resource-sharing, and upcycling factories.

Lean PLM departs from conventional front-loading design by avoiding excessive early-stage decisions and enabling progressive optimization across design, process, equipment, and operations. It integrates the entire engineering chain—from product design to factory operations—while incorporating feedback from the shop floor to achieve continuous adaptation and improvement.

The framework consists of three-layer architecture of function, information, and data supported by a data model integrating BOM (Bill of Materials), BOP (Bill of Process), and BOA (Bill of Assets). The content of over 131 meta-data for this framework has already been defined and preliminarily reviewed by 30 IVI Task Group member companies comprising manufacturing users and system suppliers, enabling early adoption.

Also, Integration Profiling, a mechanism provided to automatically match each entity’s own definitions, rather than enforcing rigid standardization, enables interoperability across distributed systems while preserving data sovereignty and leveraging the unique characteristics of each entity.

Furthermore, Lean PLM complements Circular PLM by bridging internal production perspectives with external user and environmental viewpoints. This dual approach enables synchronization between short product development cycles and longer equipment lifecycles, supporting a resilient, sustainable, and competitive manufacturing ecosystem.

April 2026
Industrial Value Chain Initiative
President, Professor Yasuyuki Nishioka, Dr-Eng

Introduction

In the face of energy issues, global environmental challenges, rare earth shortages, supply chain disruptions, and a sharp decline in the workforce, the situation surrounding manufacturing can only be described as severe. On the technological front, the emergence of generative AI is dramatically reshaping the division of roles between humans and systems, with many intellectual tasks once performed exclusively by humans now being replaced by systems. Manufacturing is undergoing significant turbulence from both internal and external forces.

In this uncertain and unpredictable environment, the manufacturing industry continues to strive for survival by making ongoing investments, exploring customer needs, and developing new products that anticipate societal changes. This paper focuses on product development that enriches society, viewing it as a model in which various goods and information circulate across society as a whole. It examines product design not as an isolated activity, but within the context of the supply chain and engineering chain, and proposes “Lean Product Lifecycle Management,” in which related functions, information, and data are interconnected and work collaboratively.

The term “lean” originated from studies of the strength of the Japanese automotive industry around 1990. “Lean production,” even before the term was coined, had been adopted by manufacturers in the latter half of the 20th century as a way to shift from mass production to high-mix,

low-volume production. Production was intentionally divided into small lots that did not benefit from economies of scale, and only the necessary quantity was produced, even when there was excess capacity. This approach, which went against conventional wisdom at the time, ultimately strengthened Japanese manufacturing.

Lean Product Lifecycle Management is characterized by avoiding over-specification at the product design stage, which serves as the starting point of product development. Instead, it preserves flexibility so that decisions can be finalized at subsequent stages, allowing for iterations and rework when necessary. In some cases, even after production preparation or actual production has begun, design changes are routinely made due to factors related to production equipment and processes. Rather than centralizing all information within the design department, the departments that form the engineering chain within the organization collaborate to progressively define both the overall structure and detailed specifications of the product.

Lean Product Lifecycle Management makes the product design process lean. In other words, it avoids deciding everything upfront during the design stage. Traditionally, based on the concept of front-loading, all information was consolidated in the early phase of design, and decisions made there were not to be changed in later stages. In reality, however, many aspects cannot be fully understood until they are actually tried and executed.

For example, one approach is to first design materials and equipment, and then design

products that can be efficiently manufactured using them. Another approach is to focus on specific usage scenarios of a product and, based on the environment and disposal processes involved, design the optimal product form and delivery method accordingly. Product design is not an art of creating something from nothing, but rather a rational and logically explainable process of finding optimal combinations under given constraints, and it should also be positioned as a learning process through which the organization can continuously evolve.

Chapter 1 presents the fundamental vision to be adopted within the evolving environment surrounding manufacturing, along with several goal scenarios tailored to the position of each manufacturer. It also introduces Lean Product Lifecycle Management as a key direction for next-generation product lifecycle management, and clarifies its role within a broader circular economy context.

Chapter 2 first reviews the current scope of product lifecycle management and then clarifies the relationships among product design, process design, and equipment design cycles within Lean Product Lifecycle Management. It also provides an overview of the functional and information models that should be considered as manufacturing standards, covering all activities related to product lifecycle management as part of the engineering chain.

Chapter 3 presents the data model for product lifecycle management. It discusses not only the product model, but also its relationships with the process model used to manufacture it and the equipment model that actually performs production. In particular, in Lean Product Lifecycle Management, the focus shifts from the product itself to the production processes that create it and the factories and equipment that support them.

To realize Lean Product Lifecycle Management, it is necessary to provide data used across diverse activities in the engineering chain in forms accessible to business applications and to establish mechanisms for distributing this data within the organization. Chapter 4 presents a method for expressing the data model required for Lean Product Lifecycle Management in a common way by aligning it with the information model.

Chapter 5 explains the mechanisms that enable multiple activities or business solutions with different backgrounds to interoperate through data. Rather than forcing individual activities to conform to a predefined common model, it presents concrete approaches for achieving integration while leveraging the unique characteristics of each activity.

This white paper is positioned as an explanatory guide to the Lean PLM Technical Specification published by IVI under the same title. For technical details, please refer to the Lean PLM Technical Specification.



Vision and Goal of Next-Generation Manufacturing

Three Visions of Manufacturing Industries

Manufacturing remains a core industry worldwide, supporting both economic activity and employment. To sustain its role in regional stability and prosperity, the sector should be guided by three key visions: **sustainability**, **inclusiveness**, and **autonomy**.

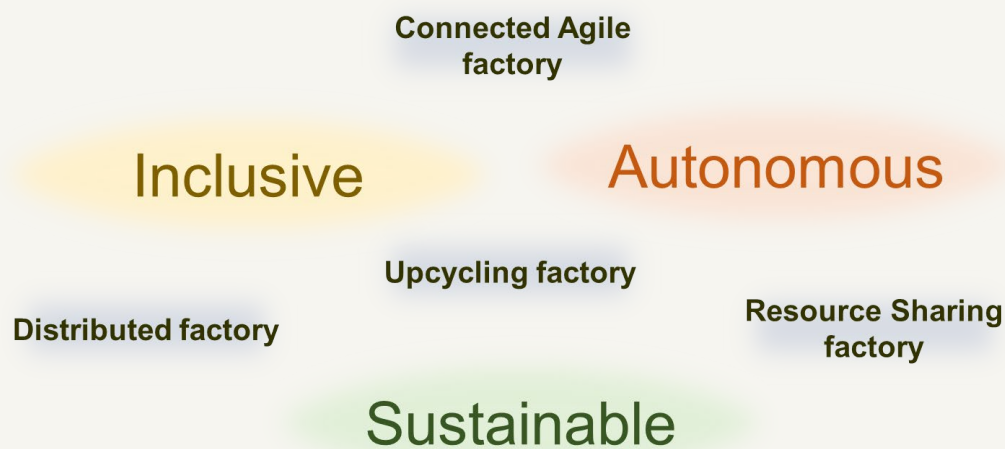


Figure 1 Three Visions of Manufacturing Industries

Sustainable

Sustainable manufacturing aims to create long-term value while minimizing environmental impact and resource consumption. It emphasizes the efficient use of energy and materials, reduction of waste, and circular approaches such as reuse, remanufacturing,

and upcycling. It also takes an integrated view of the entire lifecycle—from design to disposal—and is supported by transparent metrics such as carbon footprint and data-driven decision-making. By balancing economic performance with environmental responsibility, it ensures competitiveness and resilience even under resource constraints and high uncertainty.

Inclusive

Inclusive manufacturing promotes the participation and value creation of diverse stakeholders, including large enterprises, SMEs, workers, and local communities. By enabling shared access to manufacturing resources, digital platforms, and standardized interfaces, it lowers barriers to entry. Supporting distributed production and collaborative ecosystems allows each participant to leverage their strengths while maintaining fair and transparent relationships. Based on human-centered design, it also enhances workers' capabilities through digital tools and knowledge sharing. This approach fosters innovation, regional revitalization, and sustainable growth through broad and balanced engagement.

Autonomous

Autonomous manufacturing does not simply mean automation or unmanned operation; rather, it refers to systems in which humans and machines collaborate, continuously learn and adapt, and perform sensing, decision-making, and execution. By leveraging AI, digital twins, and real-time data, production processes are optimized and performance is improved while dynamically responding to change. Autonomy extends beyond individual machines and requires system-level coordination across processes and networks. By empowering the shop floor and enabling faster decision-making even for highly specific challenges, it achieves high efficiency and resilience in complex and uncertain environments.



Four Factory Models

Manufacturers that strongly embrace these three visions tend to converge into four types of factories, depending on their position within the supply and engineering chains, the scale of their business, their relationships with partners, and their core competencies and business models.

Connected Agile Factory

A connected factory (agile factory) leverages digital technologies such as IoT and cloud computing to enable fast and flexible production, collaborating with external partners to address challenges that cannot be solved independently. By connecting equipment, processes, and stakeholders across the entire value chain, it can respond quickly to demand fluctuations and product diversification. It emphasizes short lead times and continuous improvement, using data-driven decision-making to enhance efficiency and quality. By strengthening the integration of design, production, and services, it supports mass customization and resilient product supply.

Resource Sharing factory

A sharing factory aggregates the manufacturing resource needs of multiple companies—including equipment, facilities, and human resources—and meets those needs through large-scale investments. By leveraging digital platforms and standardized interfaces, it enables dynamic allocation of production capacity based on demand, thereby improving equipment utilization. Aggregation also facilitates the accumulation of manufacturing know-how. For SMEs, it reduces the burden of initial investment and encourages new product development.

Distributed factory

A “factory anywhere” (convenience factory) is a distributed factory model composed of multiple small-scale production sites located close to demand. Replacing centralized mass production, it integrates and coordinates these distributed sites through a digital platform. This approach reduces transportation costs, shortens lead times, and improves responsiveness to local needs. By sharing standardized data models and processes and ensuring interoperability and scalability, it achieves both global optimization and local flexibility.

Upcycling factory

An upcycling factory transforms used products and by-products into new products with higher value. Unlike conventional recycling, which typically results in a reduction of product value, upcycling enhances value by improving functionality and design. This model embeds the circular economy into manufacturing, reducing waste and maximizing resource efficiency. By combining advanced processing and regeneration technologies, it achieves both environmental impact reduction and the creation of new revenue opportunities.

These factory types are not limited to advanced firms; as conditions evolve, companies of all sizes will adopt elements of them. These factory types are not limited to advanced firms; as conditions evolve, companies of all sizes will adopt elements of them.

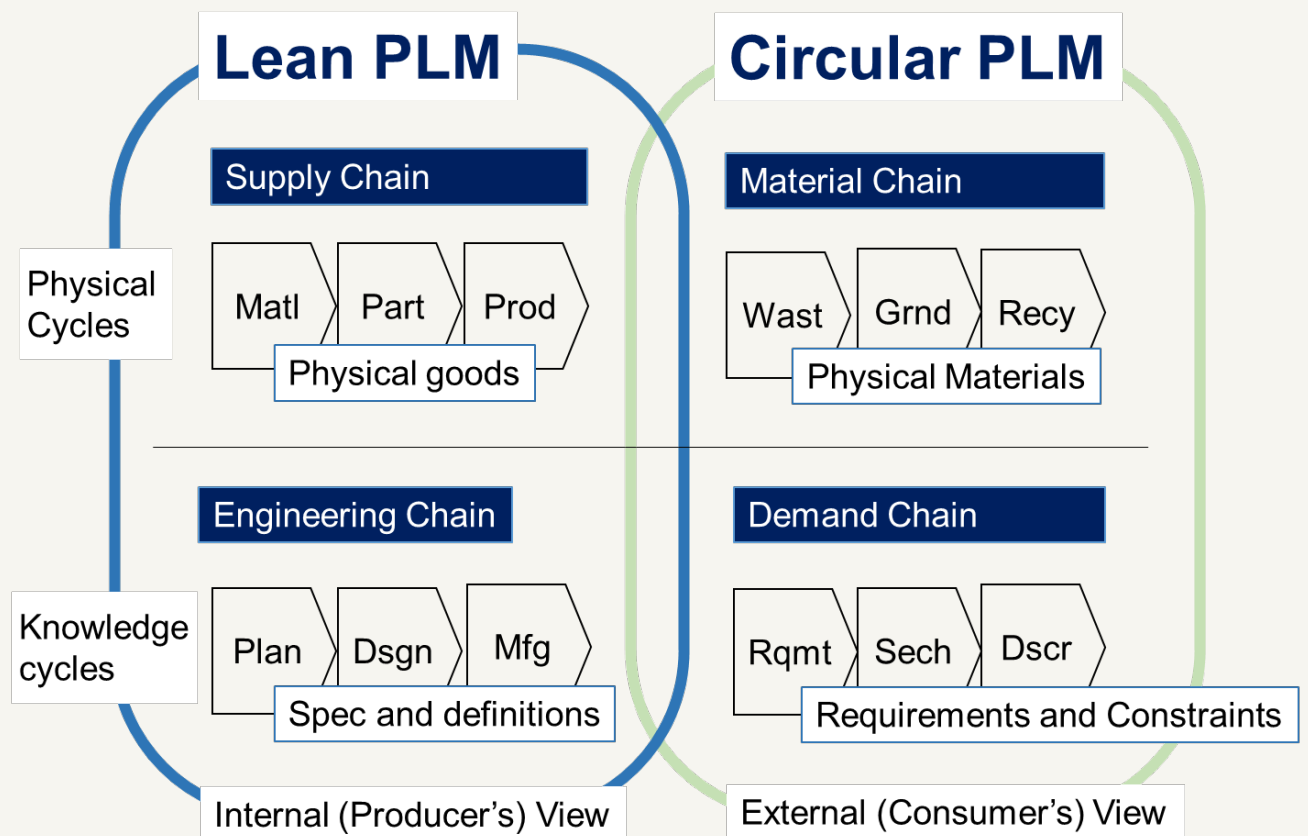


Figure 2 Positioning of Lean Product Lifecycle Management



Lean PLM and Circular PLM

As we enter a new era, when considering product lifecycle management within the context of the circular economy, there exists not only a product lifecycle viewed from the perspective of those who create products, but also a lifecycle seen from the perspective of those who use and ultimately dispose of them.

First, we organize the product lifecycle from the perspective of producers within the enterprise. Here, the product cycle refers to a sequence in which a product begins as raw materials, is processed into components in the factory, and is ultimately assembled and shipped as a finished product. This entire flow constitutes part of the physical cycle within the supply chain.

On the other hand, when focusing on knowledge and information related to a product, the product is first defined through product planning, then goes through design and manufacturing, and is finally delivered to consumers. Eventually, the product reaches the end of its sales life, but this entire sequence can also be viewed as a cycle. In general, when referring to product lifecycle management, it is this cycle that is meant.

Next, consider the product from the perspective of users outside the enterprise. After purchase, a product that is no longer used for some reason may be discarded, or it may be reused for other purposes or by other users. It may also be disassembled and reconfigured as part of another product, or crushed and recycled as material resources. These processes form part of the physical cycle as a material chain.

From the user's perspective, there is a need to be addressed even before obtaining a product. The user searches for a product that satisfies this need, discovers a suitable option, and then

proceeds to use it. When the requirements are no longer met, the product is discarded and replaced by another. However, the requirements and their underlying constraints change over time; eventually, the original need may disappear, and new needs emerge. This forms a demand chain.

In manufacturing, we define Lean Product Lifecycle Management as the synchronization of the engineering chain—viewed from the internal perspective of producers—with the supply chain as the physical flow of goods, while also incorporating the perspective of external users and environmental constraints.

In contrast, we define Circular Product Lifecycle Management as a framework that examines the demand chain and material chain from the perspective of external users and global environmental considerations—such as carbon neutrality and nature positivity—while also incorporating the producer's perspective.

In this way, next-generation product lifecycle management consists of two types—Lean Product Lifecycle Management and Circular Product Lifecycle Management. Each functions autonomously, while at the level of regions and society as a whole, they operate in coordination and collaboration with one another.

For diverse decision-making entities with differing values and interests to contribute to overall societal sustainability and inclusiveness through their autonomous activities, it is essential to establish a digital data integration platform and to ensure the preservation of data sovereignty for all participating economic actors. The following chapters will further explore Lean Product Lifecycle Management from the perspective of manufacturing.



What is Lean Product Lifecycle Management?

Various Cycles Surrounding Manufacturing

In quality management activities within manufacturing, plan (P), do (D), check (C), and action (A) are repeatedly carried out as the PDCA cycle. The completion of one task drives organizational change, which in turn leads to the initiation of new tasks to address emerging challenges. By managing what would otherwise be like activities as cycles, organizations can achieve continuous growth and further evolution at both the shop floor and enterprise levels.

Let us first organize the various cycles surrounding manufacturing. To begin with, there is the fundamental cycle in which a manufacturing

company is established and ultimately concludes its role through closure or other means. Although this varies depending on economic conditions, historical context, and regional circumstances, it typically spans a long cycle of over 30 years.

On the other hand, there is a cycle of building new factories in response to business expansion or the launch of new ventures. This typically spans around ten years. In addition, there is a process configuration cycle, in which new production lines are established by utilizing existing factory floors. This cycle is shorter, averaging approximately five to ten years. This cycle is shorter, averaging approximately five to ten years.

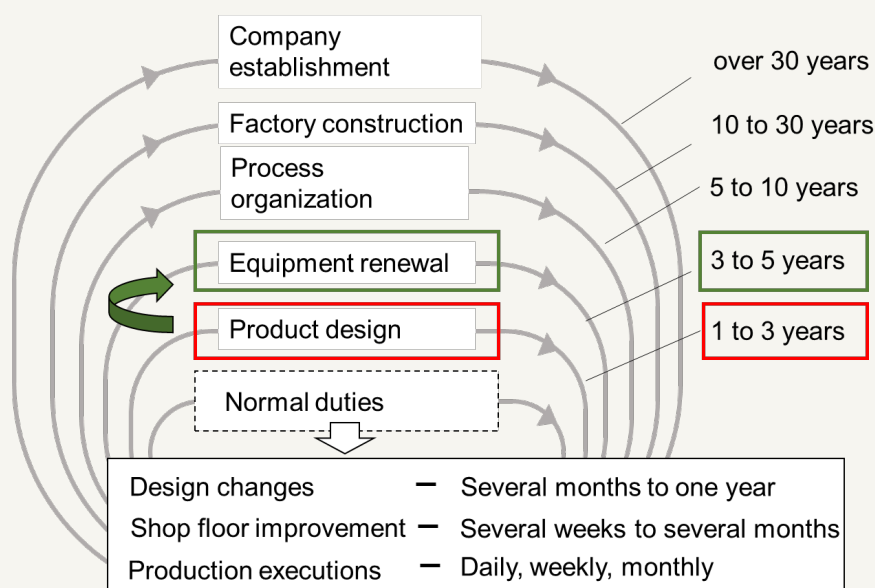


Figure 3 Various Cycles Surrounding Manufacturing

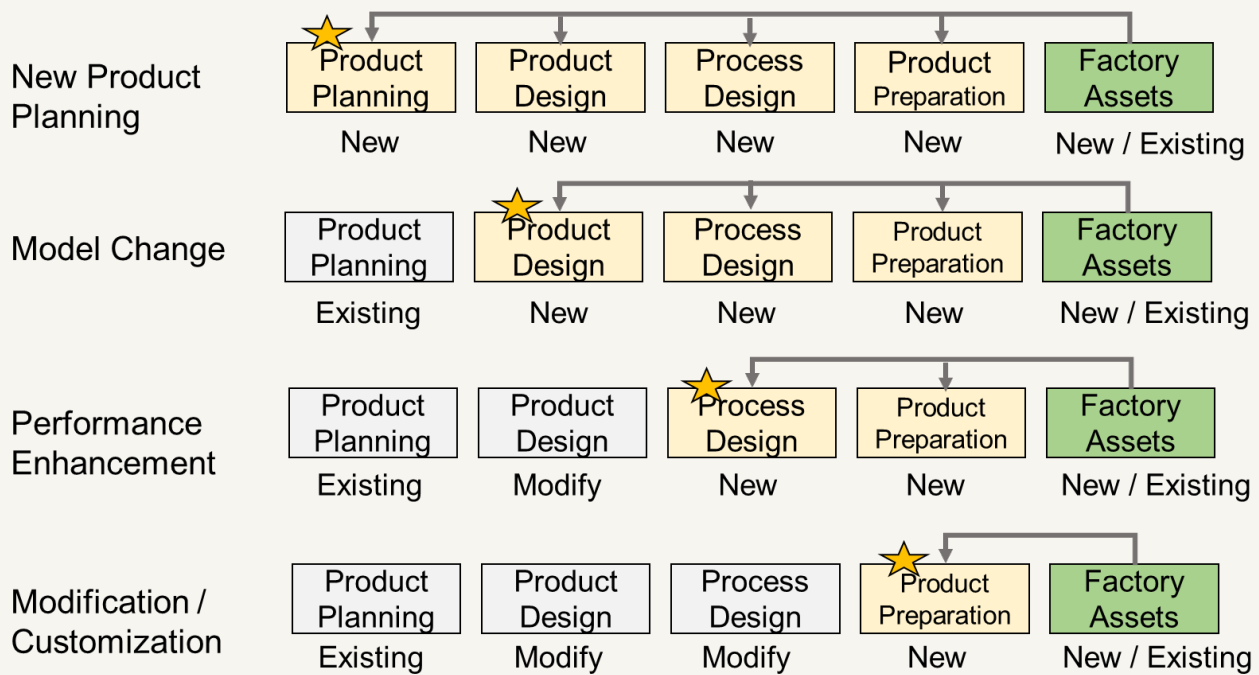


Figure 4 Design Considering Factory Asset Constraints

In new process configurations, equipment upgrades typically accompany the changes; however, even without process reconfiguration, individual equipment is periodically renewed. Although this depends on a company's performance and market trends for its products, it can generally be considered to occur on a cycle of about three to five years. These relationships are illustrated in the figure. It should be noted that the intervals of these cycles are only relative.

Compared to these equipment-side cycles - that is, the cycles on the production side—the typical product development cycle is extremely short. Although it depends on market size and technological maturity, new products are generally released on a cycle of one to three years. Product Lifecycle Management aims to further shorten this cycle.

A key challenge arises when product design requires new equipment upgrades or new process configurations. In practice, it is often the case that equipment design, equipment renewal, or process configuration is carried out with each new product

design. In reality, however, technical and economic constraints associated with equipment upgrades may prevent product design from being executed at the optimal timing, resulting in missed opportunities for market entry.

The importance of Lean Product Lifecycle Management lies in synchronizing the product design cycle with the equipment renewal cycle while maintaining their respective time-based relationships. In other words, it aims to provide technical variations within the shorter product design cycle without altering the economically optimal intervals for equipment renewal and process configuration.

In Figure 3, the product design cycle is further broken down into design change cycles, and within factory operations, into the shop-floor improvement cycle based on PDCA, as well as the actual production execution cycle. At the factory floor—where the engineering chain and supply chain intersect—it is necessary to manage and control these various cycles.

Product Lifecycle Management Driven by Equipment Design

In product development—or activities similar to product development—there are cases where a completely new product is planned from scratch, as well as various levels such as model changes of existing products, partial improvements, defect corrections, and customization. Figure 4 classifies product development into four patterns based on the flow from product planning, product design, and process design to production preparation, indicating where development actually starts anew. For example, in the case of a model change, development begins from product design based on an existing product plan, whereas in improvement or performance enhancement, the product design is largely retained and

development starts from process design. Even in cases considered “existing,” partial modifications are likely to be involved.

What is critical here is to reflect the actual conditions of production equipment and production lines, as well as future expansion plans, in new product planning, product design, and subsequent process design activities. In other words, in new product design, it is important to incorporate equipment constraints in advance while ensuring that these constraints do not unnecessarily limit design flexibility. Lean Product Lifecycle Management derives its value precisely from addressing this challenge.

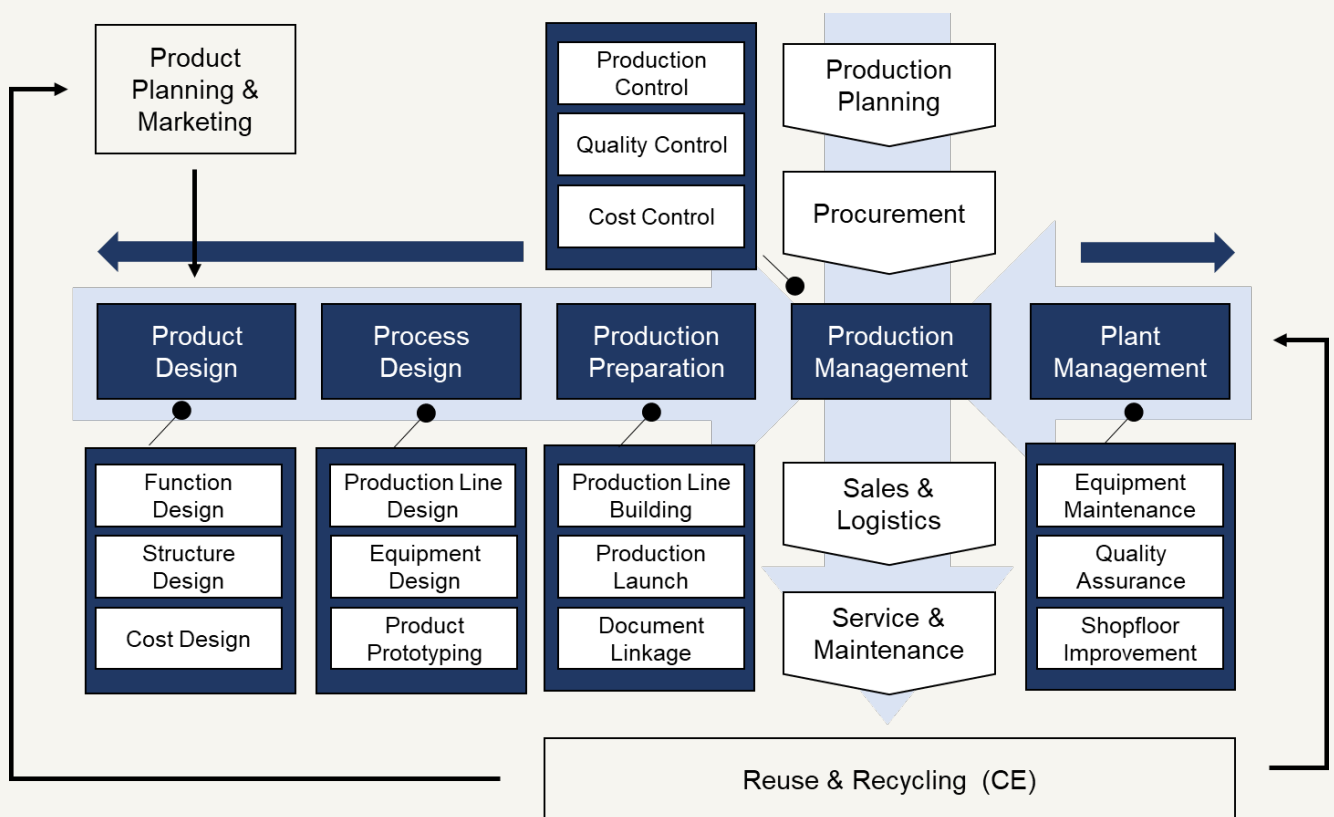


Figure 5 Functional Model in Lean Product Lifecycle Management

Fundamental Elements of Product Development

This section explains the functions that can be defined as standards for Lean Product Lifecycle Management. When product lifecycle management is viewed from an internal enterprise perspective, as shown in Figure 5, product development progresses through a sequence starting with product planning and marketing, followed by product design, process design, and production preparation. In addition to this engineering chain perspective, there is also a supply chain perspective, which includes production planning, procurement, production control, sales and logistics, and maintenance and service. Furthermore, from the standpoint of managing factory assets and enhancing added value, daily factory operations are continuously carried out.

Conventional Product Lifecycle Management mainly focused on product design, process design, and production preparation. Lean Product Lifecycle

Management extends this scope by adding functions from the production execution phase—such as production control and factory operations—resulting in five categories. In particular, by utilizing information obtained from equipment maintenance and quality assurance in factory operations within product and equipment design, it becomes possible to achieve integrated product development that unifies design and manufacturing on the shop floor.

To realize Lean Product Lifecycle Management, each activity must be described in a way that is understandable to third parties and be interoperable at the functional, information, and data levels. For this purpose, the standard model defined in the Lean PLM Technical Specification serves as a useful reference. Figure 5 shows the five functional categories and fifteen functional groups that constitute the functional model for Lean Product Lifecycle Management defined therein.

Product Design

Product design defines product specifications based on customer requirements and manages structured product information centered on the Bill of Materials (BOM). It assumes integration with manufacturing and services, while ensuring change management and traceability.

Process Design

Process design defines the process structure required to manufacture a product and standardizes and organizes it as a Bill of Process (BOP). It designs work procedures and equipment conditions, and establishes optimal production methods while maintaining alignment with product design.

Production Preparation

In production preparation, production resources such as equipment, jigs, and personnel are

arranged based on the process design. Factory assets are modeled using the Bill of Assets (BOA), and productivity and quality are ensured through pre-launch verification and simulation.

Production Management

Production control develops production plans based on demand and manages progress, quality, and inventory in an integrated manner. By leveraging real-time data, it visualizes gaps between plans and actual results and supports rapid decision-making.

Plant Management

Factory operations optimize the overall operation of the factory, including equipment, energy, and human resources. Through maintenance and continuous improvement activities, they maximize asset value and achieve ongoing efficiency improvements and sustainable factory management.

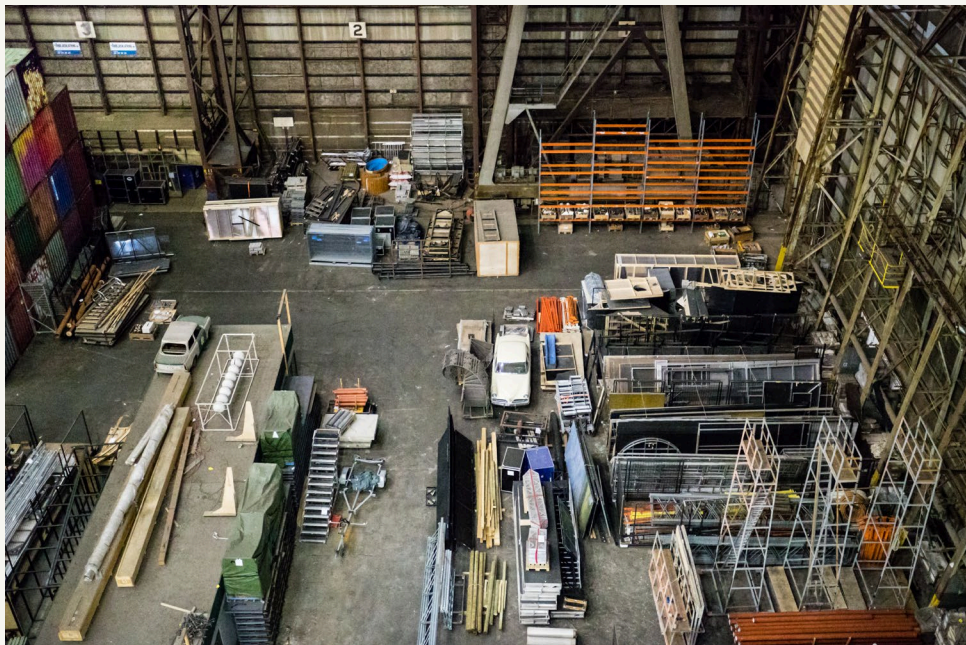
Table 1 Function Model

Category	Group	Description
Product Design	Functional Design	defines required product functions and performance based on customer needs, translating them into clear specifications that guide subsequent design, engineering, and manufacturing activities.
	Structural Design	designs product structure, including components, layout, dimensions, and materials, to realize defined functions while ensuring manufacturability, reliability, and cost effectiveness.
	Cost Design	establishes target costs and embeds cost considerations into design by optimizing materials, processes, and specifications to achieve profitability throughout the product lifecycle.
Process Design	Production Line Design	designs production lines by defining process flow, equipment layout, and task allocation, optimizing productivity, throughput, and operational efficiency.
	Equipment Design	defines specifications, structure, and control requirements of manufacturing equipment, ensuring safety, performance, and compatibility with production processes and operational conditions.
	Product Prototyping	builds prototypes to validate design assumptions, evaluate functionality and performance, and identify issues for improvement before mass production.
Production Preparation	Production Line Building	constructs and installs production lines and equipment based on design specifications, ensuring readiness and alignment with planned production processes.
	Production Launch	stabilizes processes and equipment during the transition to mass production, ensuring quality, productivity, and consistent output under real operating conditions.
	Document Linkage	integrates design drawings, specifications, and work instructions, ensuring consistent and synchronized information sharing across engineering, production, and related functions.
Production Management	Production Control	monitors and controls production processes, tracking progress and performance while managing deviations to ensure alignment with production plans.
	Quality Control	monitors product and process quality, prevents defects, and implements corrective actions to continuously improve quality performance.
	Cost Management	tracks and analyzes production costs, comparing planned and actual values to identify improvement opportunities and enhance profitability.
Factory Operation	Equipment Maintenance	performs preventive and corrective maintenance to ensure equipment reliability, minimize downtime, and sustain stable production operations.
	Quality Assurance	ensures compliance with quality standards and customer requirements through systematic processes, audits, and validation activities.
	Shopfloor Improvement	identifies inefficiencies and issues on the shopfloor, driving continuous improvement through process optimization and workforce engagement.

Table 1 describes the functional groups belonging to each functional category. In the technical specification, ten specific use cases are presented as elemental functions for each of the functional groups listed here. As an example, Table 2 below shows ten elemental functions related to the functional group “Functional Design.”

Table 2 Description of the Elemental Group for “Functional Design”

No.	Functional Element	Description
1	Product Structure Design	Designs the structure, geometry, and modular configuration required to realize product functions, taking into account manufacturability and maintainability.
2	Definition of Component Configuration	Defines the components and units that constitute the product and determines the optimal configuration, including the use of standard and common parts.
3	Creation of CAD Models	Creates 3D models of products and components using CAD, and utilizes them for design verification, interference checking, and digital prototyping.
4	Creation of Design BOM	Organizes component structures, specifications, and quantities at the design stage, and registers them as an Engineering BOM (E-BOM) in the PLM system to manage design information.
5	Detailed Product Design	Specifies materials, dimensions, tolerances, and strength, and produces detailed design drawings suitable for manufacturing, assembly, and inspection.
6	CAE Analysis of Product	Performs analyses such as structural, fluid, thermal, and vibration analysis to quantitatively evaluate performance, strength, and reliability, and to optimize the design.
7	Failure Mode Analysis	Analyzes failure modes using methods such as Design FMEA and FTA, identifies causes and effects, and develops measures to improve reliability.
8	Design Change Management	Manages design changes occurring during prototyping, evaluation, and production stages, analyzes their impact, and ensures proper implementation and approval.
9	Collaboration with Partner Companies	Shares data with component suppliers and external design partners to optimize design accuracy, lead time, and cost.
10	Compliance with Destination Specifications and Regulations	Verifies design and manufacturing conditions in accordance with safety, environmental, and certification regulations in each country or region, and reflects them in product specifications.



Standard Information Model

Each functional element in the functional model uses or provides certain information within the processes that realize that function. The information required to implement Lean Product Lifecycle Management is defined as an information model in the Lean PLM Technical Specification. Similar to the functional model, the information model consists of five categories and fifteen groups. The five information categories are as follows.

Business Planning Information

It is information used to define the direction of products and business, integrating plans, specifications, and costs to clarify objectives and constraints. It supports decision-making in upstream phases and serves as the foundation for development and production activities.

Development Information

It is information used to concretize products and processes, systematically defining structures, procedures, and criteria. It standardizes design intent and forms the technical foundation for ensuring quality and productivity.

Engineering Information

It is information that translates designs and plans into executable forms on the shop floor and enables coordination across departments and

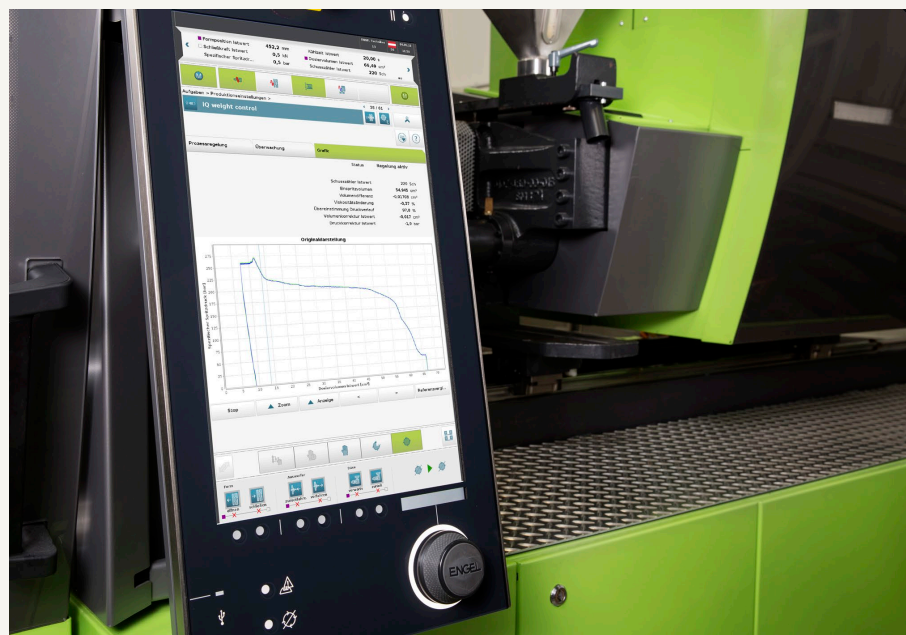
systems. It includes quality information and ensures consistency and executability of products and processes.

Management Information

It is information used to control the progress of production and operations, integrating schedules, instructions, and work content. It maintains alignment between planning and execution and supports efficient operations and rapid decision-making.

Shop-Floor Information

It is information used to understand shop-floor conditions and drive improvement, including 4M, performance, and improvement data. Through problem visualization and root cause analysis, it supports continuous improvement and the enhancement of operational capabilities.



Business Planning Information	Development Information	Engineering Information	Management Information	Shopfloor Information
Planning Information Development Plan Equipment Planning Sheet	Structural Information Bill of Materials (BOM) Bill of Process (BOP)	Deployment Information Failure Mode and Effects Analysis (FMEA) Quality Function Deployment (QFD)	Schedule Information Master Production Schedule (MPS) Work Breakdown Structure (WBS)	4M Information Production Control Sheet Skill Management Sheet
Specification Information Product Design Specification Process Design Specification	Procedural Information Quality Control Process Chart Production Procedure	Consolidation Information Value Stream Map (VSM) Traceability Information	Instruction Information Production Order Gantt Chart	Status Information Production Progress Board (Andon) Inventory Management Sheet
Cost Information Standard Cost Sheet Actual Cost Sheet	Control Criteria Information Quality Control Criteria Sheet Environmental and Safety Criteria Sheet	Quality Information Product Inspection Sheet Equipment Inspection Sheet	Performance Information Production Performance Sheet Equipment Performance Sheet	Improvement Information Issue Management Sheet Defect Management Sheet

Figure 6 Classification and Examples of the Information Model

Table 3 provides an overview of the information groups that further subdivide each information category. In practice, structured information—such as computer screens and documents—is identified and utilized by names such as screen titles or document names.

For example, the “Engineering Bill of Materials (E-BOM)” and the “Bill of Assets (BOA)” belong to the information group “Structure Information” within the information category “Development

Information.” Similarly, the “QC Process Chart” belongs to the information group “Procedural Information,” and “Process FMEA” belongs to the information group “Deployment Information.”

The Lean PLM Technical Specification defines 56 information models as standard models. Representative examples are shown in Figure 6, categorized by information category and information group. For details of each information model, please refer to the technical specification.

Table 3 Description of the Information Model

Category	Group	Description
Business Planning Information	Planning Information	Information that defines product development and production objectives, including methods, quantities, timelines, and resource allocation, serving as the basis for coordinating overall activities.
	Specification Information	Information that defines the functions, performance, and conditions of products and processes, and is shared as the basis for design, manufacturing, and quality.
	Cost Information	Information that manages cost structures, estimates, and actuals related to products and processes, and is used for profitability analysis and improvement activities.
Development Information	Structural Information	Information that defines the components and relationships of products and equipment, serving as the foundation for structural management such as BOM and BOA.
	Procedural Information	Information that defines the specific methods and sequence of operations and processes, contributing to the establishment of standard work and the stabilization of quality.
	Control Criteria Information	Information that defines criteria and specifications related to quality, operations, and equipment, serving as a common basis for evaluation and management.
Engineering Information	Deployment Information	Information used to translate design and planning content into executable forms across processes and departments, enabling practical implementation.
	Consolidation Information	Information used to link and share data across departments and systems, ensuring data consistency and interoperability.
	Quality Information	Information that includes measurement values and evaluation results related to product and process quality, serving as the foundation for defect prevention and quality improvement.
Management Information	Schedule Information	Information that indicates the start and end times and progress of production and operations, used for scheduling and progress control.
	Instruction Information	Information that specifies detailed instructions for operations and production, directly supporting execution on the shop floor.
	Performance Information	Information that represents actual activity results—such as production quantities, work outcomes, and operational status—and serves as the basis for analysis and improvement.
Shopfloor Information	4M Information	Information related to workers, equipment, materials, and methods, used for factor management and related purposes.
	Status Information	Information that enables real-time monitoring of equipment and process conditions and anomalies, supporting stable operations and rapid response.
	Improvement Information	Information that includes root cause analysis of issues and defects, as well as countermeasures, and is used for continuous improvement and prevention of recurrence.

Basic Structure of a Value-Connecting Data Model

Three-Layer Architecture

Lean Product Lifecycle Management adopts a three-layer architecture consisting of a functional model, an information model, and a data model to describe the actual business activities of each manufacturing enterprise. This architecture is not limited to Lean Product Lifecycle Management; it can also be applied to Circular Product Lifecycle Management and other scopes of challenges. The following sections explain the functional model, information model, and data model.

Function Model

Lean Product Lifecycle Management adopts a three-layer architecture consisting of a functional model, an information model, and a data model to describe the actual business activities of each manufacturing enterprise. This architecture is not limited to Lean Product Lifecycle Management; it can also be applied to Circular Product Lifecycle Management and other scopes of challenges. The following sections explain the functional model, information model, and data model.

Information Model

A model that defines what information is required and what information is produced to realize each function in business activities. It is presented in a form that supports decision-making by operators or responsible personnel and serves as a unit that leads to specific actions. It exists in physical contexts through specific media, such as computer screens and documents.

Data Model

A structured representation of the content that composes information, including specific values such as numbers and text displayed in fields. It exists independently of the corresponding information. Data described according to the data model can be processed based on defined logic and transmitted to different locations using communication mechanisms.

When discussing the efficiency of business operations through the use of data, the focus may be on either actual data that is referenced and manipulated in day-to-day operations, or on metadata that describes the structure and procedures of that data. While the term “data” generally refers to actual data, a “data model” refers to the latter—metadata—and is used for purposes such as information system design and analysis, as well as the development of mechanisms for data integration.

BOM, BOP, and BOA

To understand the relationship between the information model and the data model, we consider the “Bill of Materials (BOM),” “Bill of Process (BOP),” and “Bill of Assets (BOA)” as examples. Within the information model, these belong to the information group called “Structure Information.” As the names suggest, they represent the structures of parts, processes, and equipment, and serve as the core elements in various forms of collaboration across the engineering chain.

Bill of Materials (BOM)

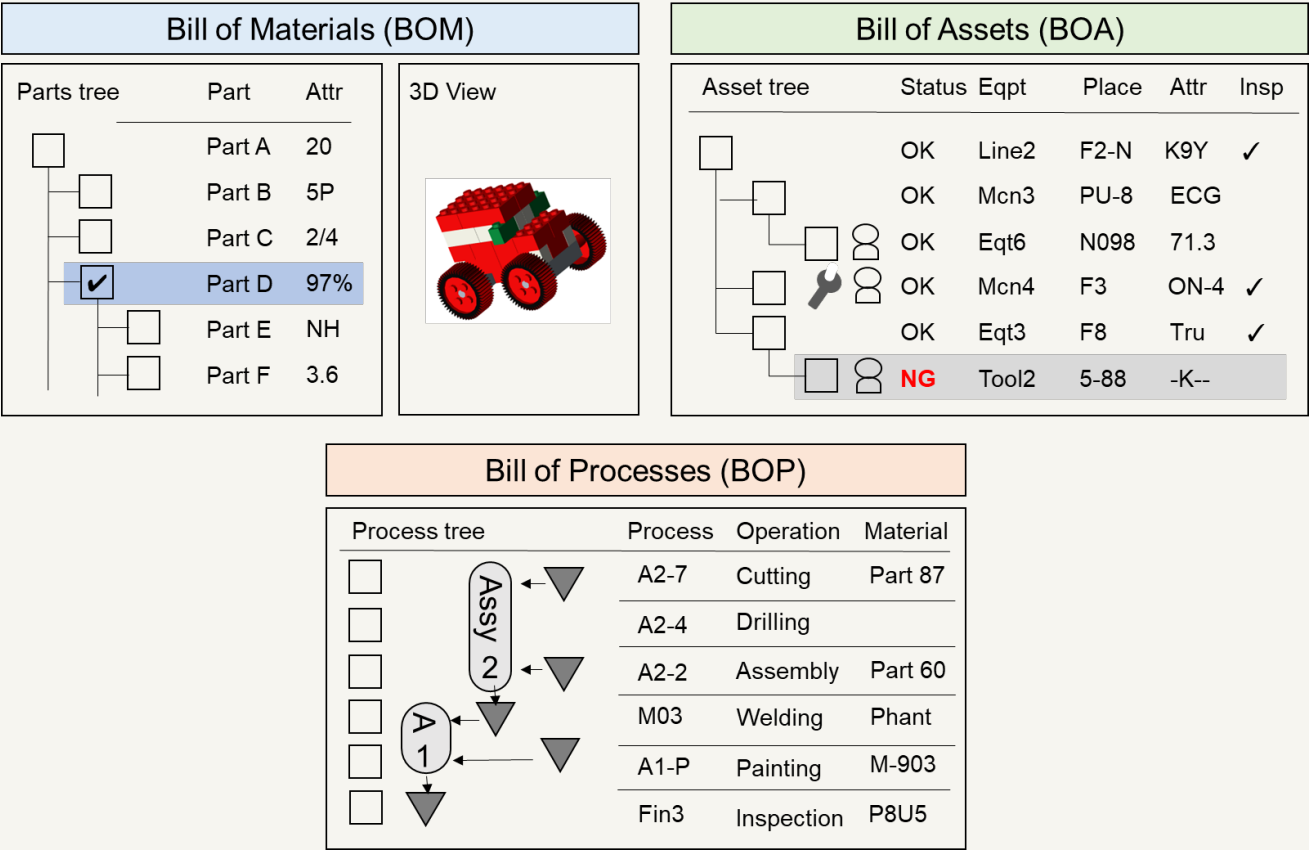
Defines the hierarchical structure and quantity relationships of the parts and materials that constitute a product. It often originates as an Engineering BOM (E-BOM) created during the design phase and is then expanded into a Manufacturing BOM (M-BOM) or Service BOM (S-BOM) according to manufacturing and service requirements. Beyond its traditional role as a parts list, the BOM is closely linked to a wide range of activities such as design change management, cost estimation, procurement, and inventory management, serving as a core element of information integration throughout the entire product lifecycle.

Bill of Process (BOP)

Defines the processes and work procedures needed to manufacture a product, including their sequence, required equipment and tools, and operating conditions. Based on BOM, production engineering uses the BOP to design actual manufacturing processes. It supports production planning, work instructions, process improvement, and quality management, forming the basis for standard operations. By linking equipment and human resources, the BOP ensures shop-floor executability and enables flexible, optimized production.

Bill of Assets (BOA)

Defines the structure and relationships of factory assets—equipment, devices, units, and workers—essentially functioning as a BOM for the entire production system. By linking equipment hierarchies, layouts, capabilities, status, and maintenance history, it supports operation management, maintenance planning, and investment decisions. Combined with the BOM and BOP, it enables integrated management of product, process, and equipment, improving visualization and optimization of the manufacturing site. Further value is expected through integration with the Asset Administration Shell (AAS).



Data Model as an Integration Platform

As shown in Figure 7, Bill of Materials (BOM), Bill of Process (BOP), and Bill of Assets (BOA) each have their own presentation forms (views), which serve as interfaces used within the functional model. Each information model is defined by the information items that compose it, and the arrangement of these items forms the corresponding information model form.

When defining an information model as a standard, essential information items are specified, while the layout is provided only as a reference and not prescribed. This is because the internal layout of information may vary across different business activities.

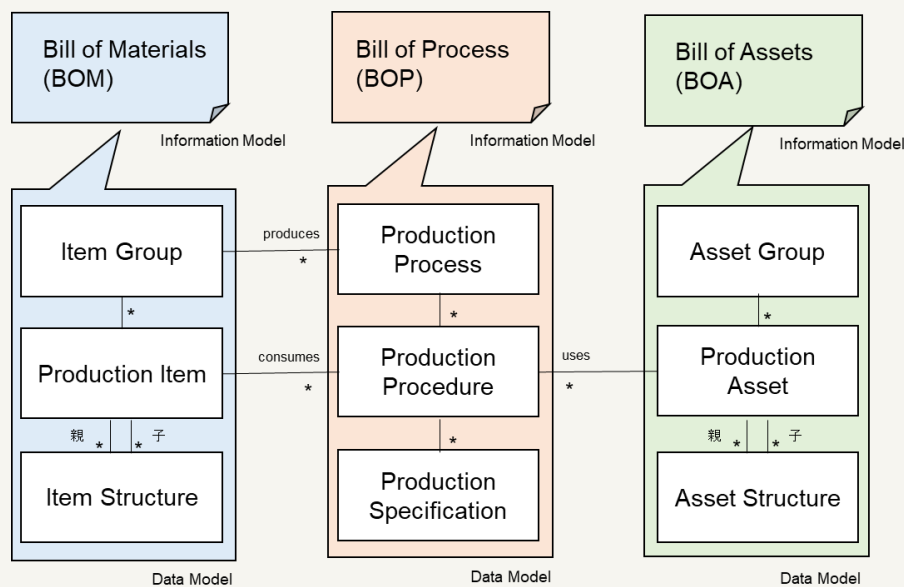


Figure 8 Relationship between the Information Model and the Data Model

Furthermore, the information model is specified in detail by the data model. As shown in figure 8, Bill of Materials (BOM), Bill of Process (BOP), and Bill of Assets (BOA) are each represented by corresponding data models and their relational structures. For example, the BOM is modeled using entities such as “Item Group,” “Production Item,” and “Item Structure.” Here, the “Item Structure” represents the parent–child relationships among items within the BOM, while the “Item Group” is a non-physical entity used to group multiple items that share common specifications. The data models corresponding to the Engineering BOM (E-BOM), Manufacturing BOM (M-BOM), and Service BOM (S-BOM) follow the same basic structure shown here.

As shown in Figure 8, Bill of Assets (BOA) has the same data model structure as the Bill of Materials (BOM). Production equipment itself is an Equipment supplier’s product. Therefore, the equipment defined in BOA can be configured by directly referencing BOM of that equipment. The assets in the BOA include all resources necessary for production, such as existing equipment, jigs, and workers.

Products represented by the Bill of Materials (BOM) and equipment or production lines represented by the Bill of Assets (BOA) are linked through the Bill of Process (BOP). In this sense, the BOP serves as the junction point of the engineering chain within the factory and can be regarded as a core entity for relating various data models.

As shown in the figure, the data model corresponding to the Bill of Process (BOP) represents “Production Process,” the “Work Procedures” that compose it, and the detailed content of the “Production Procedures” through “Production Items.” In terms of the relational structure among entities, the “Production Process” indicates the “Item Group” to be produced, and within the “Production Procedure,” it specifies the entities corresponding to the “Production Items” to be consumed and the “Production Equipment” to be used.

In this way, the data model defines, within the implementation of the corresponding information systems, the information items that compose the information model as well as the relationships among information models. In particular, the handling of structural information with multiple hierarchical structures—such as Bill of Materials (BOM) and Bill of Process (BOP)—varies depending on the type of products and each company’s historical background. However, when these are redefined as data models as described above, such differences can often be absorbed to a certain extent, enabling them to be represented in a common structure.



Integration Profiling : Framework for Integration in Distributed Environments

Data Integration Among Stakeholders

To coordinate activities in Lean Product Lifecycle Management through data, the data models used in each activity must be identical or at least mappable to one another. In practice, this requires that the applications supporting these activities be interoperable through the data they use.

Interoperability applies not only to systems already in operation but also to software products provided by external IT vendors. Both operational systems and vendor solutions have their own

functional, information, and data models for the activities they support.

Because each business unit and IT solution maintains its own internally consistent model, it is difficult to impose external changes. These units therefore function as stakeholders in system integration. In Lean PLM, stakeholders may be actual manufacturing activities—each with its own functional, information, and data models—or external IT solutions developed for those activities.

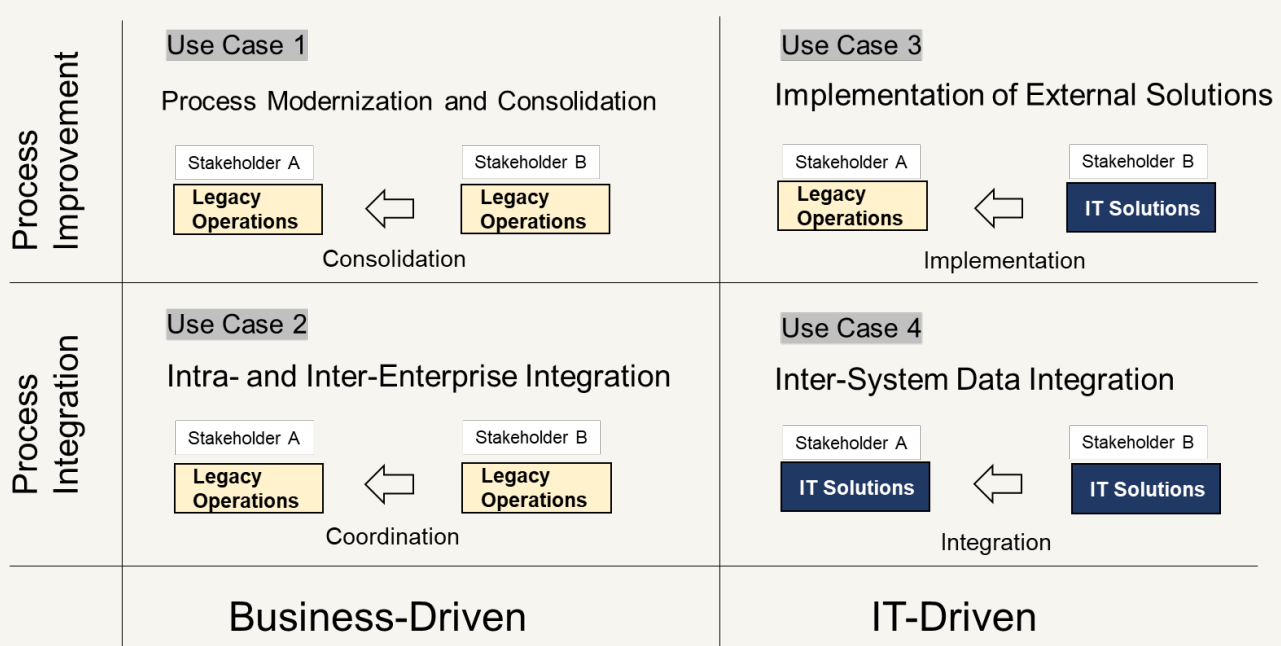


Figure 9 Use Case Classification by Stakeholder Type

Figure 9 organizes the interaction patterns between the two types of stakeholders defined above into four use cases.

Use Case 1 addresses scenarios such as the renewal of information systems within an enterprise or the integration of multiple departments. Use Case 2 covers cases where different business functions within an enterprise, or across enterprises, are interconnected; the same applies even in inter-company contexts. Use

Case 3 involves applying solutions developed by external companies to actual manufacturing operations. While this corresponds to activities such as fit-and-gap analysis, it also extends to implementation following the analysis. Finally, Use Case 4 corresponds to situations in which multiple business systems provided as external solutions are designed to interoperate with one another. In the following, these four use cases are discussed as issues of integration between two stakeholders.

Approach to Mediating a Common Model

When enabling data integration between stakeholders, it is necessary to transform data formats and content to match each party's specifications. However, it is often difficult to prepare all the information required to develop the necessary conversion programs for every potential integration partner.

Therefore, in Lean Product Lifecycle Management, a standardized format is defined in advance to describe data mappings and transformation procedures with respect to the standard model specified in the technical specification, independent of any specific

integration partner. Each stakeholder defines its content according to this method and registers it as an integration profile. By using this information, it becomes possible to automatically generate the logic required for data transformation across multiple stakeholders.

In data transformation, auxiliary entities or attributes are added as needed, and logic such as data mapping and aggregation between attributes is executed. In addition, handling for missing data and responses to errors during transformation are defined in advance.

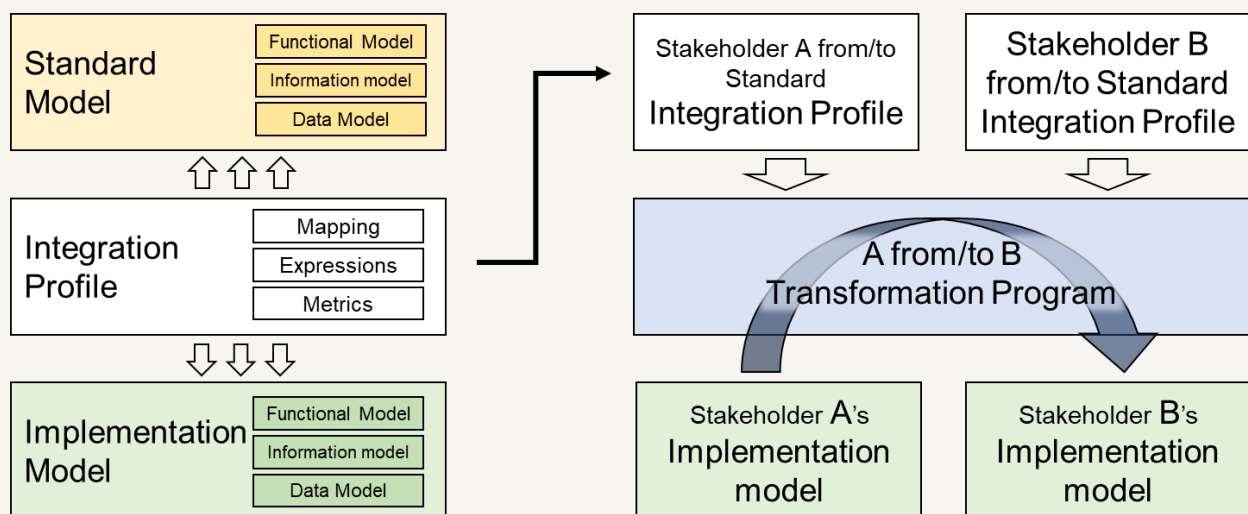


Figure10 Mechanism of Integration Profiles

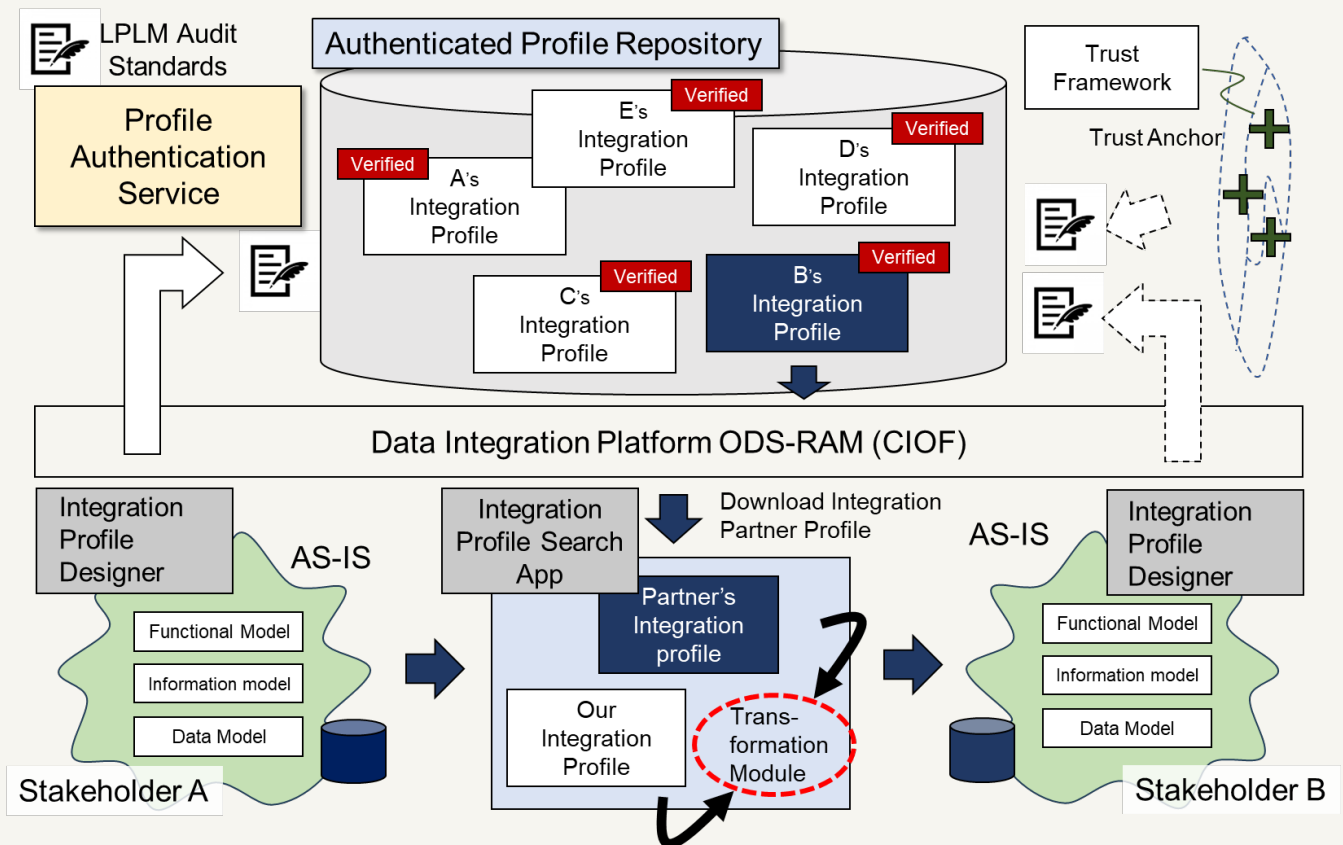


Figure 11 Mechanism for the Distribution of Integration Profiles

To actually transform data for an integration partner, it is necessary to obtain the partner's integration profile in addition to one's own, combine them to create a transformation program, and implement it as an integration module. Alternatively, the transformation module may be

omitted for the time being, and data transformation can be performed interactively based on the integration profiles, enabling execution at a minimal level without significant cost.

Mechanism of Integration Profiles

For Lean Product Lifecycle Management to contribute not only to the growth of individual companies but also to the autonomy, inclusiveness, and sustainability of entire industries, it depends on whether a wide range of stakeholders can register the details of their business activities as integration profiles and make them available for distribution as needed.

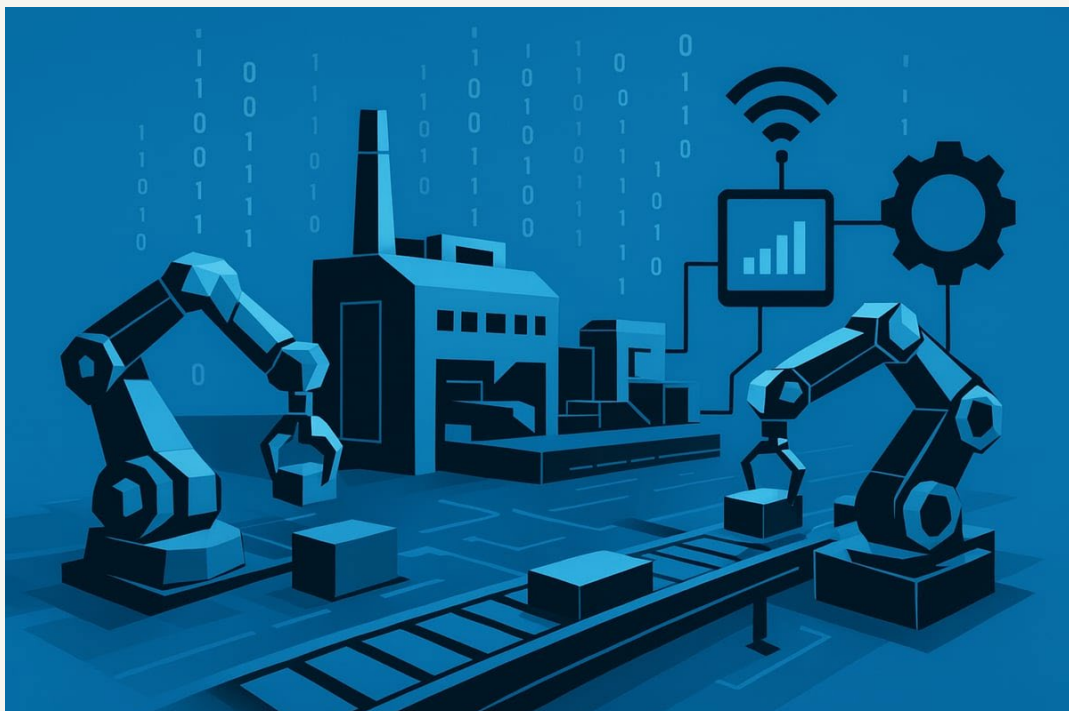
In recent years, frameworks such as Manufacturing-X in Europe and ODS-RAM (Open

Data Space Reference Architecture Model) in Japan have been proposed, and efforts are underway to establish infrastructures for trusted data integration. IVI has developed CIOF (Connected Industries Open Framework) and operates a system for selective data disclosure based on bilateral contractual agreements. By leveraging these mechanisms, manufacturing companies can collaborate with external parties while keeping sensitive production information

confidential and disclosing only the minimum necessary information.

Figure 11 shows the basic architecture of the data integration platform for Lean Product Lifecycle Management. The key point is that the data circulated on this platform is not actual operational data, but metadata in the form of definition information. While such definition information may still contain content that

companies wish to keep confidential, this concern can be addressed by disclosing it to partners under NDA through CIOF. IVI plans to provide, by spring 2027 on the ODS-RAM (CIOF) platform, both a service for certifying integration profiles for Lean Product Lifecycle Management and a mechanism to search for integration profiles of trading partners and provide them with the consent of the registered parties.



Conclusion

This paper has argued that, in the new era, it is essential for manufacturing to advance Lean Product Lifecycle Management in order to realize an autonomous, inclusive, and sustainable society. It has proposed the necessary concepts and approaches, as well as practical and effective methods for improving interoperability between systems through data integration.

As concluding remarks, we revisit the benefits of implementing Lean Product Lifecycle Management for manufacturing, organizing them from a managerial perspective.

- **Bidirectional engineering chain to shorten development cycles**

By integrating shopfloor asset structures (BOA) with operational and maintenance data into PLM, the entire equipment lifecycle—from installation to operation and renewal—can be managed consistently. This enables data-driven renewal decisions and predictive maintenance, optimizing capital investment and improving utilization. Furthermore, by maximizing value at the equipment level rather than the product level, it contributes to improved long-term asset efficiency.

- **Shopfloor-driven PLM to optimize equipment lifecycles**

By shifting from conventional design-driven approaches to product design based on production processes (BOP) and equipment constraints, the gap between design and manufacturing can be minimized. In addition to incorporating manufacturability and quality considerations from the early stages, detailed changes identified later can also be accommodated without rework. This shortens the lead time to mass production ramp-up and improves both the speed and reliability of overall development.

- **Stronger engineering collaboration with suppliers**

By systematically handling BOM, BOP, and BOA within a data model, design and manufacturing information can be shared with suppliers, eliminating inefficiencies in distributed production structures. Changes in specifications or processes can be communicated quickly, reducing coordination costs and enabling early detection of quality issues. As a result, collaborative engineering across the entire supply chain can be achieved.

- Enabling bottom-up software development platform

By developing and integrating software functions in small, shopfloor-driven units based on operational challenges and improvement activities, it becomes possible to build systems that closely reflect actual conditions. Compared with top-down large-scale development, this approach offers greater flexibility and enables rapid response to change. In addition, standardized data models facilitate seamless integration across different systems.

- Data acquisition and utilization to maximize the value of AI

By structuring and storing shopfloor technologies and skills as searchable data, tacit knowledge can be transformed into explicit knowledge. This facilitates skill transfer and reuse,

promoting knowledge utilization across the organization. Furthermore, by analyzing and leveraging accumulated data with AI, advanced decision support—such as deriving optimal conditions and detecting anomalies—can be achieved.

Traditional manufacturing has strengthened its competitiveness through advanced quality management and productivity improvements driven by continuous improvement (kaizen). In this sense, it is highly robust in the “how.” On the other hand, the “what”—what to produce, namely product design—still tends to rely heavily on the strengths of individual companies or individuals. What is needed is a system or foundation that enables product design itself to generate competitive advantage in an integrated manner and contribute directly to profitability. By reinforcing mechanisms that excel in the “how” while also establishing frameworks to win in the “what,” Lean Product Lifecycle Management is expected to further drive the advancement of manufacturing.

